

Tutorial - Week 1

1. Solve the inequalities in terms of intervals and illustrate the solution set on the real number line.

(a) $x(x - 4) \geq -2$,

(b) $\frac{x^2}{2} < 2x - 1$,

(c) $x^3 - 3x + 2 > 0$

(d) $\frac{1}{25} \leq \frac{1}{x^2 - 6x + 9}$

(e) $\frac{1}{|9 - 2x^2|} \geq \frac{1}{3}$

(f) $|\frac{3}{x} + 1| < 4$

2. Show that if $x < y$, then $x < \frac{x+y}{2} < y$.

3. Consider the line $l(x) = 3x + 2$. Prove that, for every $\varepsilon > 0$, if the distance from x to 1 is smaller than $\frac{\varepsilon}{3}$; then, the distance from $l(x)$ to 5 is less than ε .

4. Decide which of the following three points lie on the same line.

(a) $(1, 1)$, $(3, 7)$ and $(-2, -8)$,

(b) $(-1, 7)$, $(2, 5)$ and $(1, 2)$.

5. Find the equation of the line through the point

(a) $(1, 2)$ that is parallel to $y = 3x + 5$,

(b) $(-3, 0)$ that is perpendicular to $y = -4x + 1$.

6. Find the line perpendicular to the line $y = -3x + 2$ that passes through the intersection of lines $y = \frac{1}{7}x - 2$ and $y = 2x - 28$.

7. Sketch the following regions in the xy -plane.

(a) $\{(x, y) \in \mathbb{R}^2 : y^2 < x + 1 < 2y + 3\}$

(b) $\{(x, y) \in \mathbb{R}^2 : |y + 4| < 3 \text{ and } \frac{x^2}{9} + \frac{(y-4)^2}{16} \leq 1\}$

(c) $\{(x, y) \in \mathbb{R}^2 : \frac{x^2}{2} \leq y \leq x^2 + (y - 4)^2\}$

(d) $\{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 2\}$

8. Sketch the bounded regions in the xy -plane enclosed by the following pairs of curves, and express them using inequalities.

(a) The line $y = \frac{1}{3}x$ and the parabola $x = y^2$.

(b) The line $x - 2y = 2$ and the parabola $y = 4 - x^2$.

(c) The hyperbola $y^2 - \frac{x^2}{4} = 1$ and the line $y = \sqrt{2}$.

9. Let θ be an angle such that $\sin \theta = \frac{2}{3}$ and $\cos \theta > 0$. Find all other trigonometric functions at θ .

10. Find all the real numbers x such that

(a) $\cos x = \frac{\sqrt{3}}{2}$.

(b) $2 \cos^2 x + 3 \sin^2 x = \frac{5}{4}$.